

COMPUTER SCIENCE: BIG
DATA AND DATA ANALYTICS

AI-Based Skin Analysis and Skin Lesion Detection

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University year : 2025-2026

Presentation plan

- 1** Introduction
- 2** Requirements Specification
- 3** Planification and systems design
- 4** Implementation
- 5** Conclusion And Perspectives

Introduction

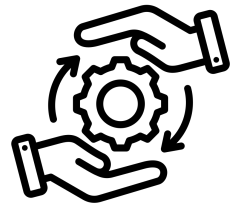


1.Context:



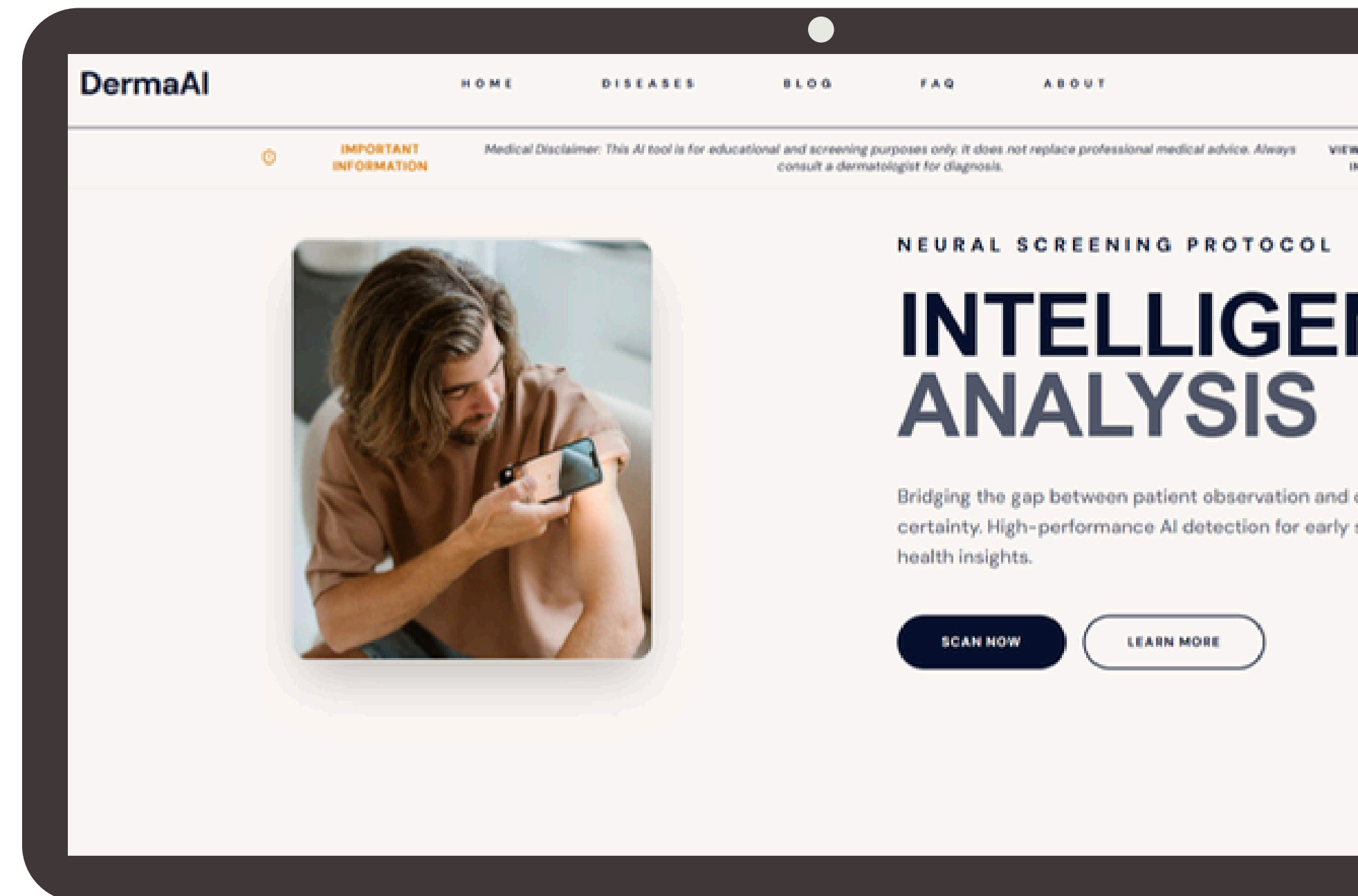
Empowering dermatological care, simplified.

Connect, scan, and track skin health with AI-driven tools.



Smarter Skin Health, for Everyone

AI-powered tools designed for patients, not just specialists.



2.Host Company:



Aura Dynamic

Tunisian agency for digital solutions

- Aura Dynamic is a digital solutions company
- Specializing in web and mobile application development, offering innovative AI-driven services like UI/UX design, business management systems , and full-cycle digital product delivery.
- They collaborated with DermaAI to enhance its visibility and user experience.

3.Problem Statement:

1



Long waiting times due to shortage of dermatologists.

2



Expensive screening fees .

3



Misleading online internet tools causing inaccurate self-diagnoses.

4



language barriers restricting access to localized regional dialects.

4.Existing Solutions :

1

Absence of explainable feedback.

2

Lack of real-time, personalized data.

3

Global platforms are expensive and not localized.

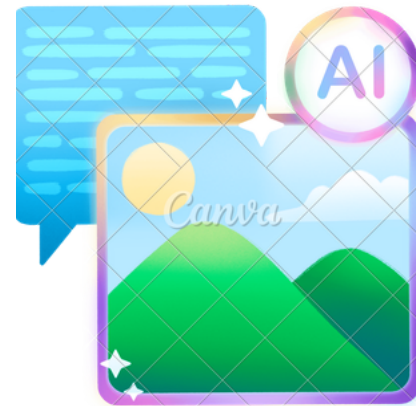
4

No Arabic language support.

5.Objectives :

- 1** | Offer skin lesion screening without requiring a specialist.
- 2** | Deliver explainable results with visual region highlights.
- 3** | Support multiple languages for diverse user groups.
- 4** | Provide scan history and report generation.
- 5** | Integrate a digital assistant for post-screening guidance.

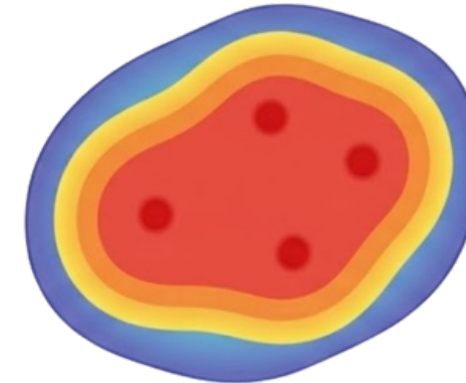
6. Proposed solution :



AI-powered skin
disease detection
via images.



Multilingual **chatbot**
for instant skin
health support.



Visual **heatmap**
highlighting AI-
analyzed skin regions.



Real-time
dermatological news
feed via web scraping.

Requirements Specification



1.Functional Requirements :

GUEST

- ✓ Register or log in securely.
- ✓ Read Educational Content on Diseases
- ✓ View Recent Dermatological News

USER

- ✓ Manage personal account
- ✓ Submit skin lesion images.
- ✓ View results with explanations.
- ✓ Track scan history .
- ✓ Interact with digital assistant.

ADMIN

- ✓ Access dashboard
- ✓ Monitor platform activity.
- ✓ Manage users and roles.
- ✓ Export data and backups.

2. Non functional Requirements :



3. *Technological Requirements :*

Frontend :



DataBase :



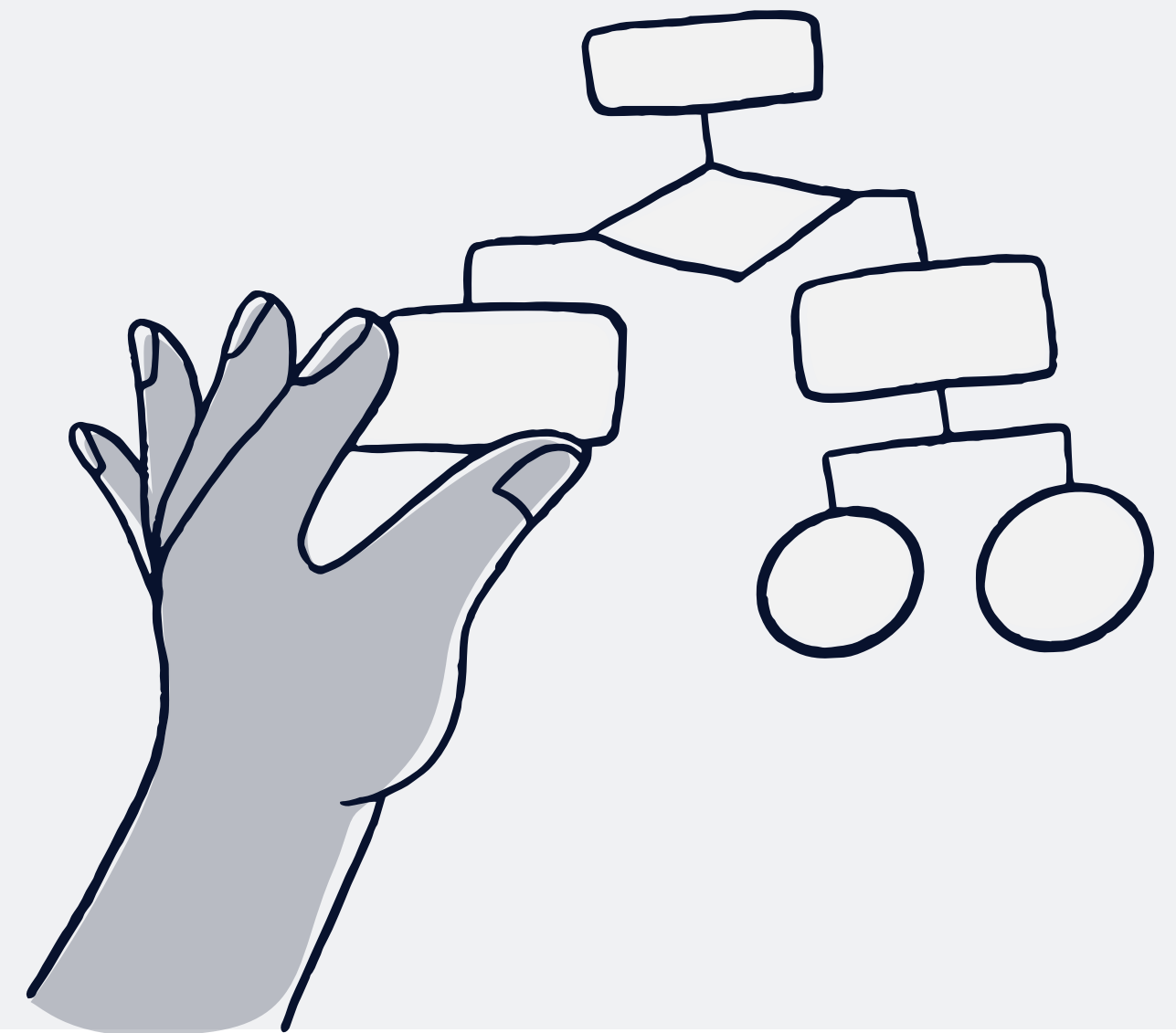
Backend :



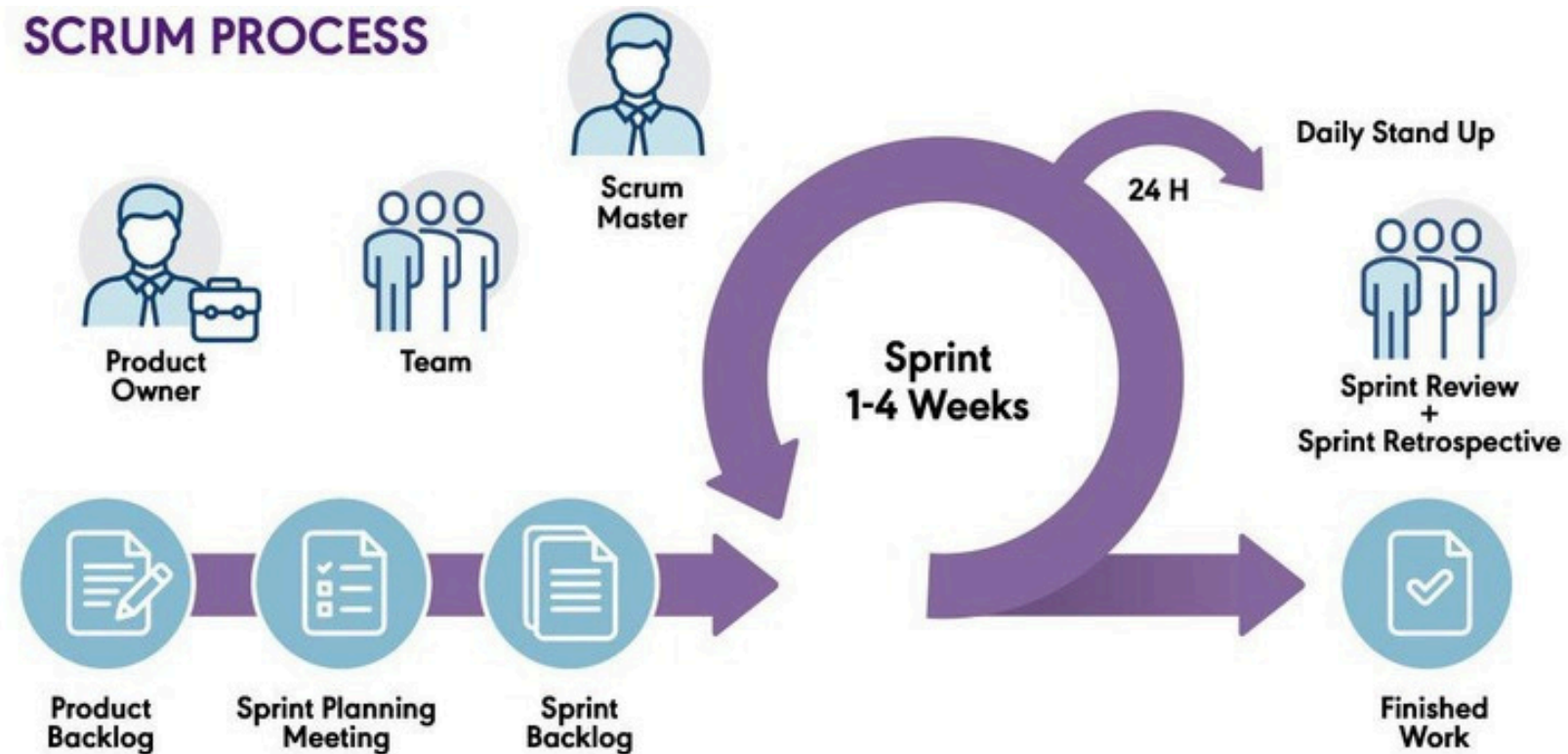
Tools :



Planification and systems design



1.Methodology: Scrum



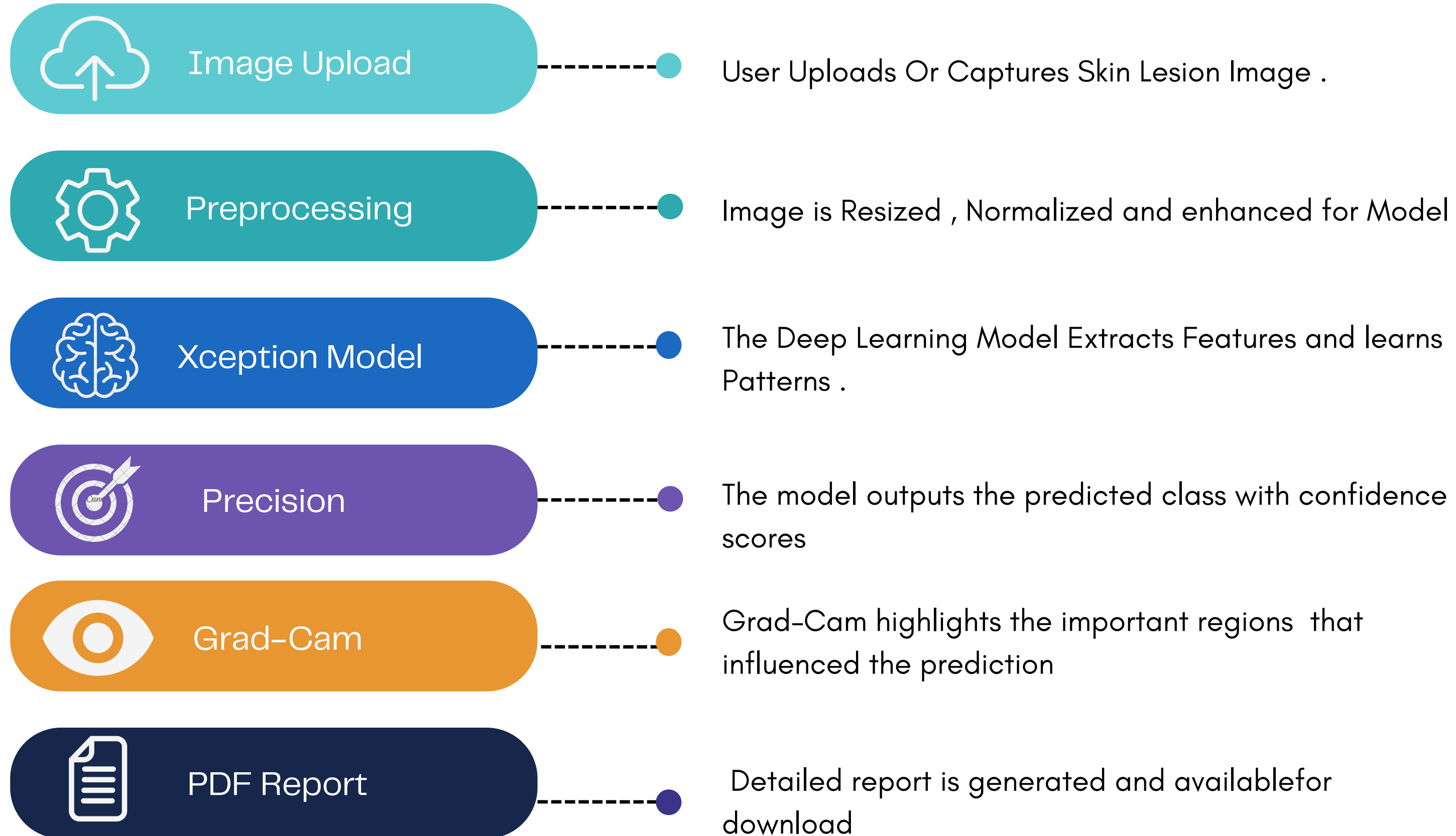
Why Scrum methodology?

- Structured sprints ensure clear timeboxing, strong team coordination, and regular progress reviews.
- Incremental delivery allows continuous feedback and easy adaptation to evolving requirements.
- Iterative development reduces risk and guarantees a functional product at every stage.

2.Sprint Implementation :



3.AI Pipeline:



Development



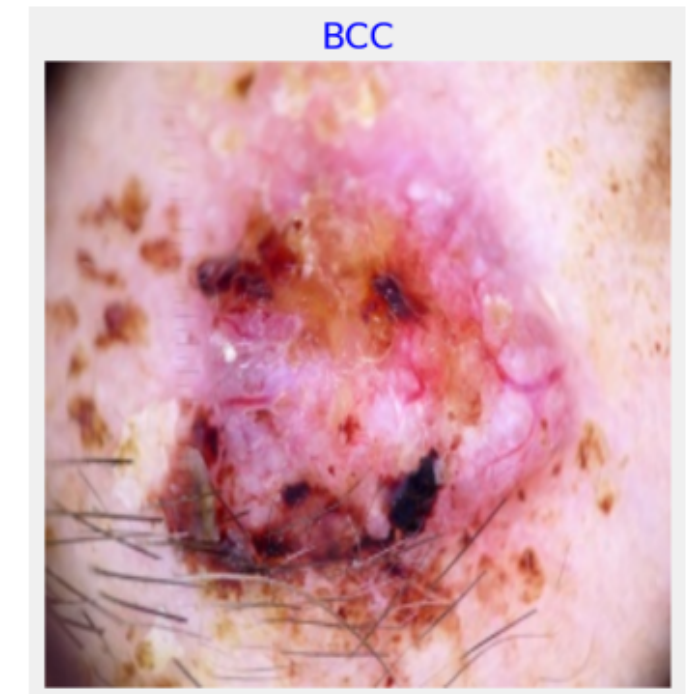
Key Features :

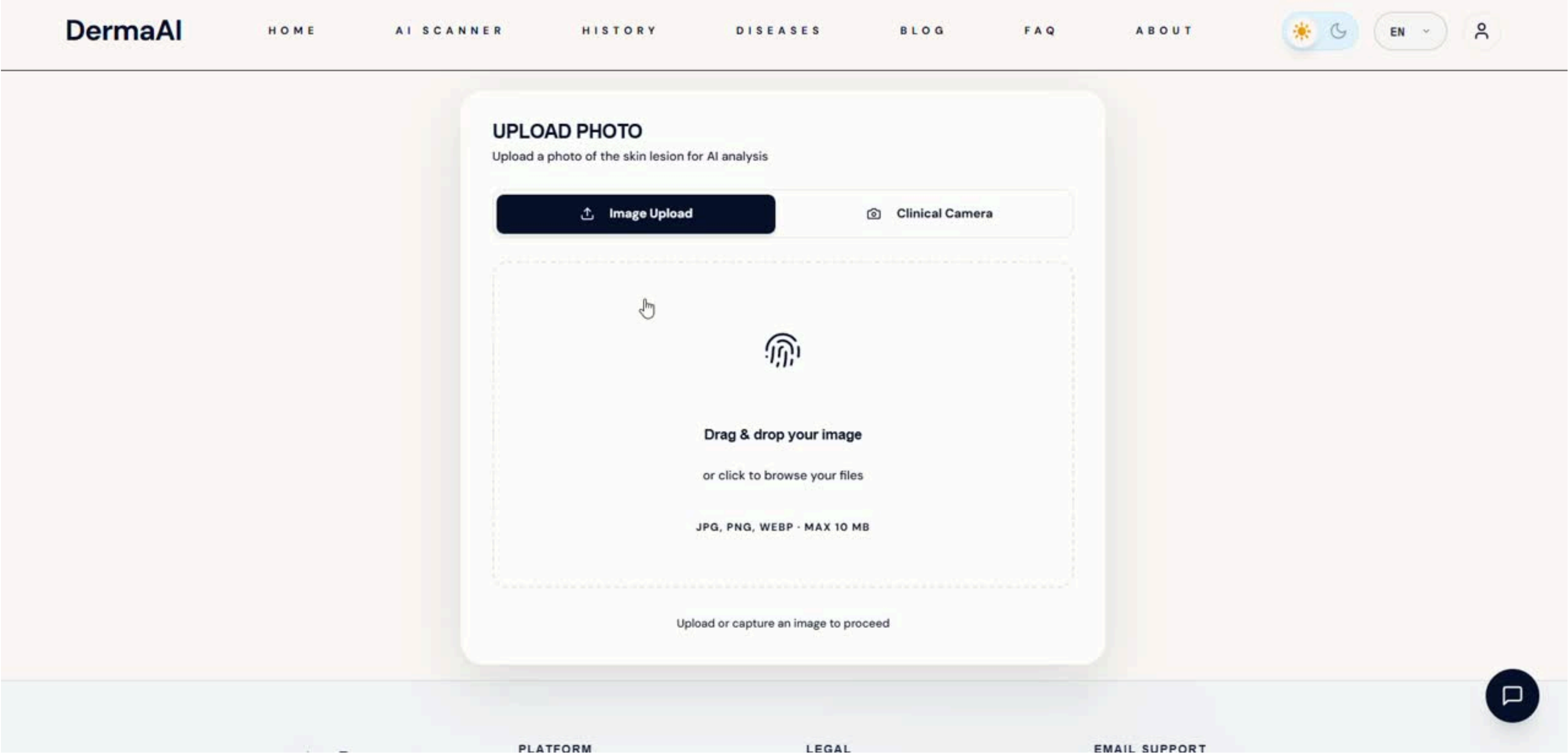
- Trained using Xception architecture (Transfer Learning).
- Integrates Grad-CAM visual heatmaps to highlight suspicious skin regions.
- Trained on Kaggle (16GB GPU) with 10000 skin disease images .
- Source: HAM10000 Dataset.
- Delivers responses in ~1 second .
- Offers over 90% confidence in predictions.

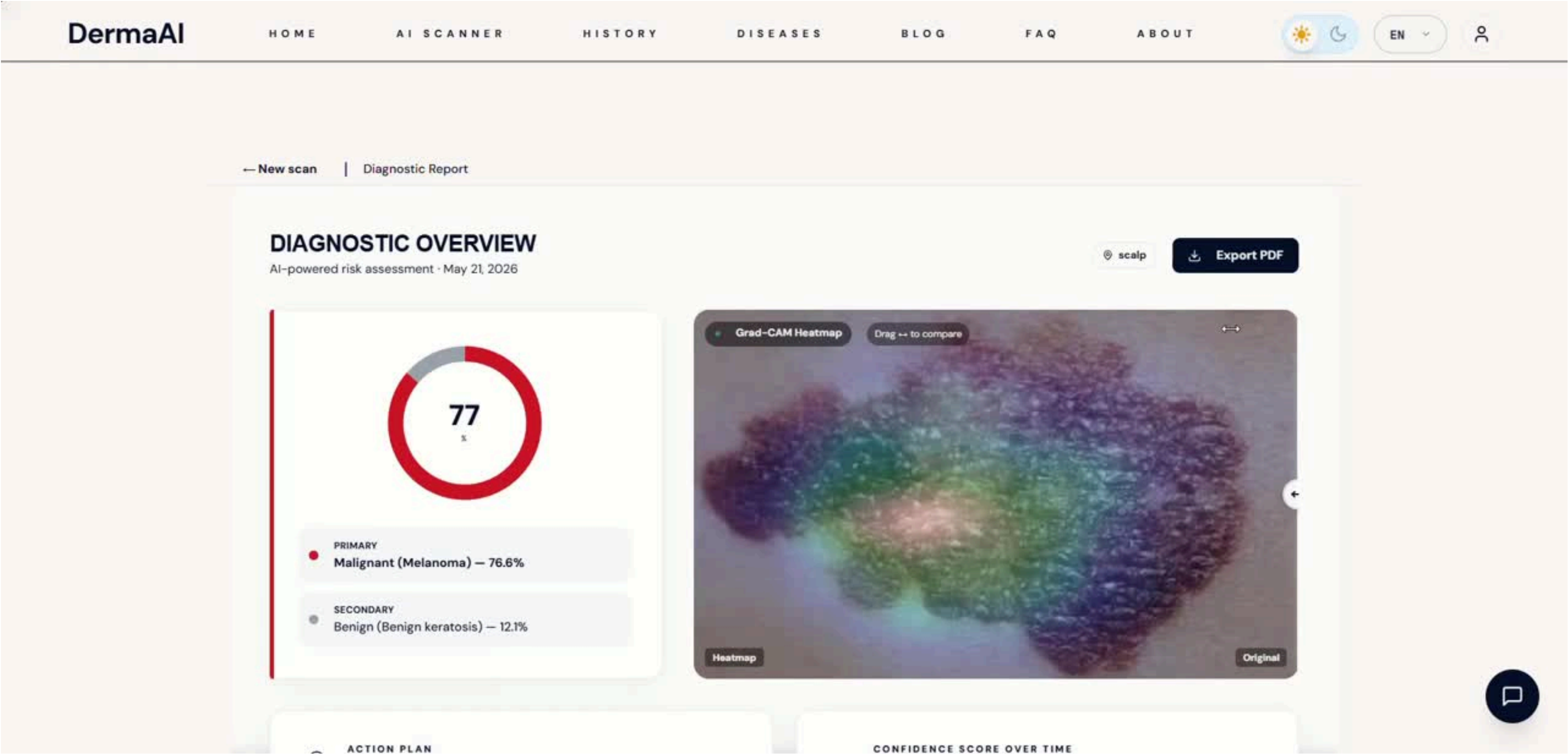
Performance Metrics :

- Accuracy: 83.8%, overall correctness.
- Precision: 75%, true positives among predicted infected skins.
- Recall: 76%, actual infected skins detected.
- F1 Score: 75%, precision-recall balance.
- Confidence: >90%, ~100% in ideal cases.
- Diagnosis Time: Optimized to run in ~1 second per scan.

```
Model is being loaded- this will take about 10 seconds  
1/1 [=====] - 1s 1s/step  
image is of class BCC with a probability of 99.79 %
```

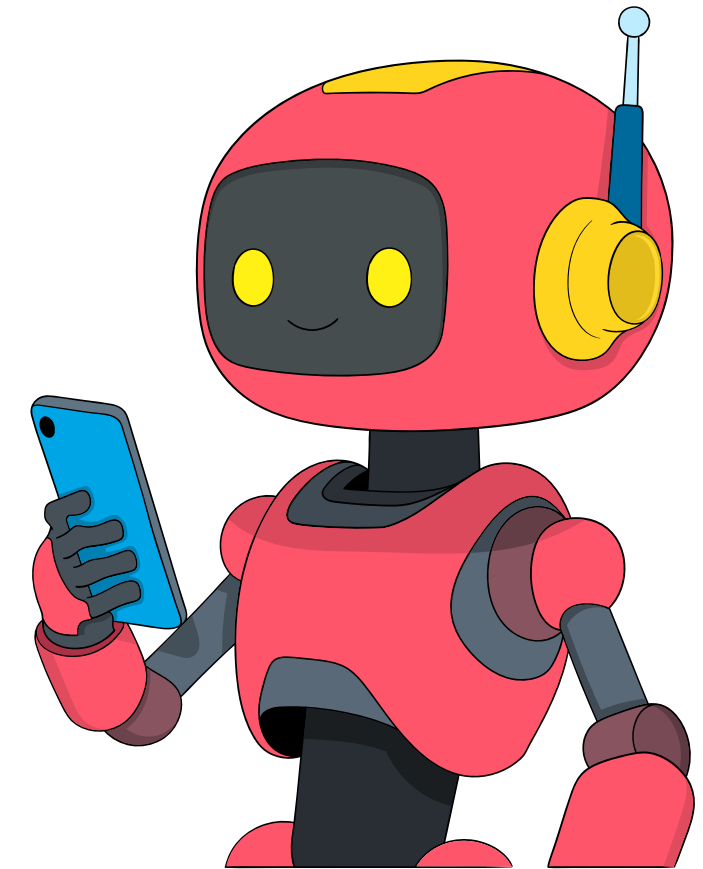






Key Features :

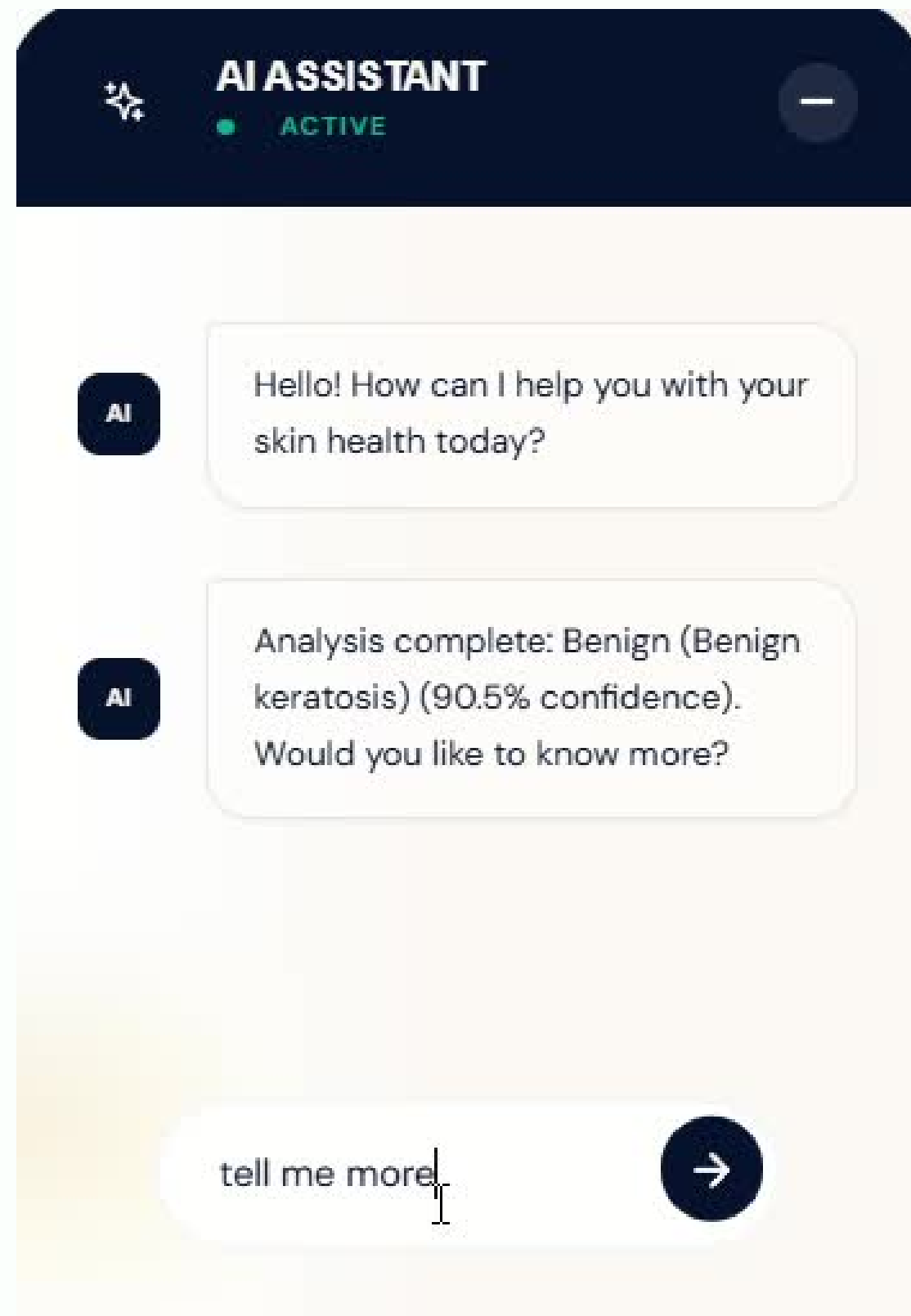
- Supports multilingual queries.
- Aware of the user's scan result and diagnosis.
- Runs fully local, no data sent externally.
- Adapts response language to the user's interface preference
- Prevents definitive diagnoses or medication advice.



AI ChatBot Model

PROOF OF COMMITMENT

English

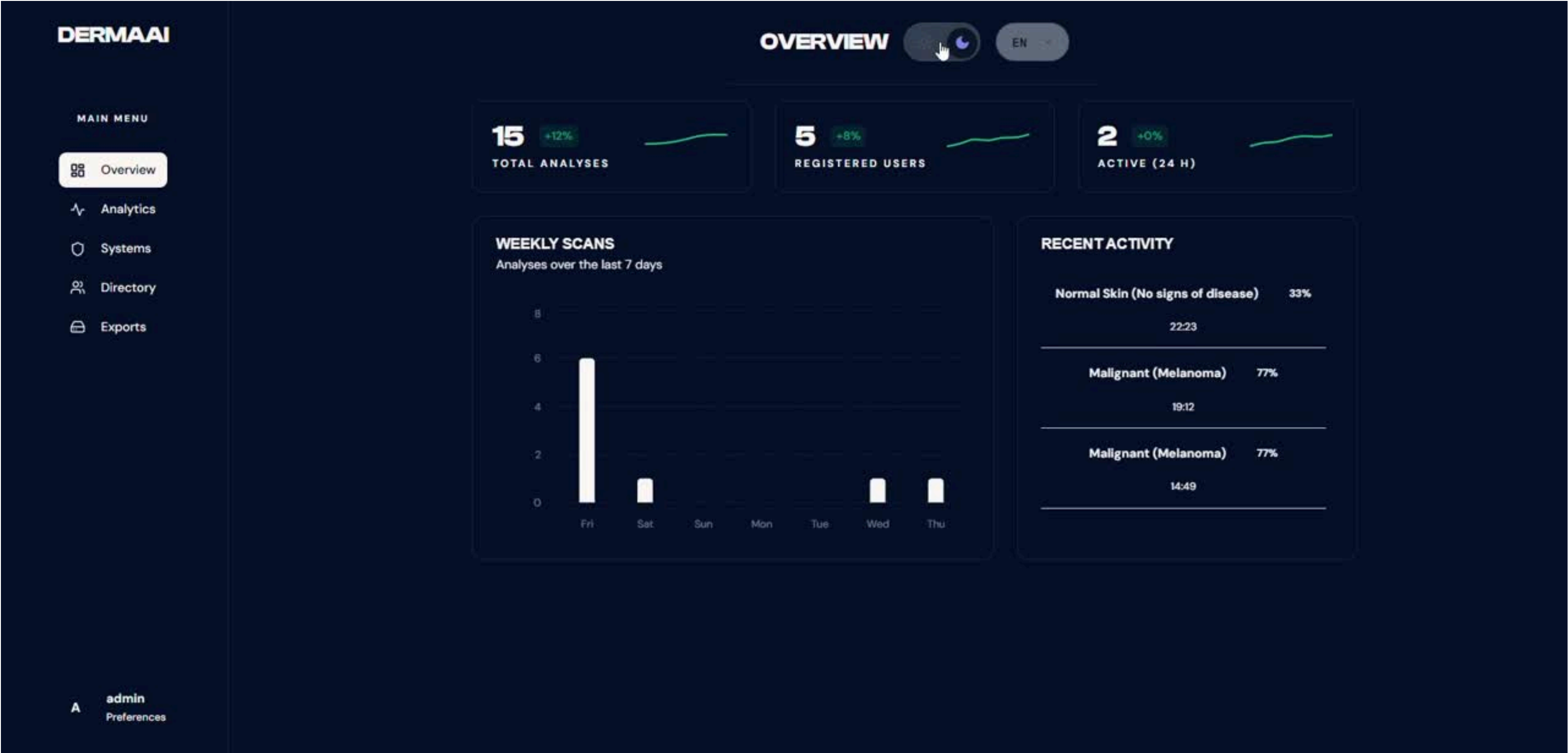


Arabic



Key Features :

- Built on React for a fast, user-friendly admin interface.
- Integrates MongoDB database for secure patient and scan log management.
- Manages system users, platform activities, and role assignments (RBAC).
- Supports real-time updates for smooth backend operations.
- Designed for secure, comprehensive platform administration.



Conclusion And Perspectives

Conclusion

- ✓ Delivered a localized, AI-driven platform for preliminary skin lesion screening.
- ✓ Enabled explainable diagnostics with Grad-CAM and scan history tracking.
- ✓ Gave Patients easy-to-use, multilingual support.
- ✓ Addressed critical dermatologist shortages and waiting times in Tunisia.

Perspectives

- ✓ Retrain models with body-selector spatial data for higher accuracy.
- ✓ Migrate local architecture to secure, scalable online servers.
- ✓ Develop dedicated iOS/Android applications for consumer access.
- ✓ Partner with hospitals and medical centers for wider impact.

THANK YOU FOR YOUR
ATTENTION!